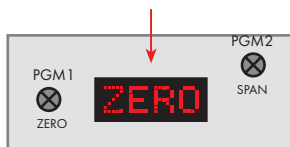


## DM-TOX



Scrolling Full Message/Text Display



(shown as PN KSA-9H7-303094-100M in SS junction box)



## Applications

- Oil & Gas
- Chemical Plants
- Food & Beverage
- Steel Mills
- Pulp and Paper
- Refineries
- Wastewater Treatment Plants
- Utilities

## Description

DM-TOX Sensors are non-intrusive "Smart" sensors designed to detect and monitor H<sub>2</sub>S in air using electrochemical sensor technology.

The intelligent plug-in, field replaceable cell provides automatic recognition of gas type and range, and features oversized gold-plated connections that help prevent corrosion.

The DM-TOX's rugged framework includes an intrinsically safe electro-polished 316 stainless steel housing with fully encapsulated electronics and dual layer surge protection.

This innovative design virtually eliminates sensor failure due to water ingress, corrosion, vibration, and transient spikes.

A primary feature of the DM-TOX is its embedded intuitive software that simplifies operator interface by guiding the user through routine calibration, configuration, and fault diagnostic functions using a built-in alpha/numeric display.

The DM-TOX is equipped with standard analog 4-20mA, and HART outputs.

Teledyne Gas and Flame Detection's H<sub>2</sub>S sensors have a long shelf life and are supported by an industry-leading warranty.

## Features

### Failsafe User-Friendly Interface

- LED Display (With Antiglare Cover)
- Full Text Display Method
- Non-intrusive Interface
- Auto Zero/Auto Span
- Pre-emptive Fault Diagnostics

### Environmentally Bulletproof

- Electropolished 316SS Construction
- 100% Epoxy Encapsulated Circuitry
- Bulletproof I/O Protection
- Water-proof, Corrosion-Proof, Vibration-Proof

### Modular and Serviceable

- Modular Design
- Plug and Play Components
- Quick Thread Release Sensor Endcap (Allen Wrench only required)
- Integral Calibration Port

## System specifications

### Sensor Type

Continuous diffusion/adsorption  
3-electrode electrochemical cell  
Plug-in field replaceable Intelligent Type

### Outputs

Linear 4-20 mA DC  
HART

### Hazardous Location Approvals

Explosion Proof & Intrinsically Safe  
cCSAus (80209826)  
Class I, Division 1, Groups ABCD  
Class I, Zone 1, Group IIC  
IECEX (DEK 24.0039X)  
Ex db ib IIC T4 Gb (Tamb = -40C to +55C)

### Ingress Protection

NEMA 4X, IP66

### Safety Approvals

cCSAus  
IECEX

\* Specifications subject to change without notice

## Order guide

PN KSA-9H7-303094-100M DM-TOX Sensor Assembly with HART and 316 SS J-Box

## Integration options

The first three characters after KSA is the Integration options as follows:  
KSA 9H7- Hart Integration Module (Hart Communication Protocol version 7.0, HART Registered)

## Sensor Specifications

|                           |                                                                   |
|---------------------------|-------------------------------------------------------------------|
| <b>Gas</b>                | Hydrogen Sulfide (DC)*                                            |
| <b>Warranty</b>           | 2 years                                                           |
| <b>Measuring Range</b>    | 0-100ppm                                                          |
| <b>Accuracy</b>           | ±10% applied gas concentration or<br>±3 ppm, whichever is greater |
| <b>Response Time</b>      | T20 ≤ 20sec<br>T50 ≤ 45 sec<br>T90 ≤ 60 sec                       |
| <b>Operating Temp.</b>    | -40 to +131 °F/-40 to +55 °C                                      |
| <b>Storage Temp.</b>      | -31 to +131 °F/-35 to +55 °C                                      |
| <b>Operating Humidity</b> | 5-90% RH non-condensing                                           |

\* Note3: DC=Desert Climate Sensors are 3rd-party test report to ISA 92.00.01 2010 standard

## Mechanical specifications

### Dimensions

11"H x 6.1"W x 3.75"D; 280mmH x 155mmW x 96mmD (with junction box)  
Mounting holes (J-box) 5.5"; 140mm center to center

### Weight

9 lbs; 4.08kg (w/stainless steel j-box)

## Electrical specifications

### Power Input

11 - 30 VDC

### Power Consumption

Normal operation = 30mA @ 24V (<0.75 watt)  
Maximum = 50mA @ 24V (1.2 watts)

### Inrush Current

500mA @ 24V (typical)

### RFI/EMI Protection

Complies with EN50270

### Analog Output

Linear 4-20mA DC (1,000 ohms max loop load @ 24VDC)  
0mA All Fault Diagnostics  
2mA In-Calibration  
4-20mA 0-100% full-scale  
22mA Over-range condition

### Status Indicators

4-digit LED display with gas concentration  
Full-script menu prompts for AutoSpan,  
Set-up Options, and Fault Reporting

### Faults Monitored

Loop, Input Voltage,  
Missing Sensor, Zero,  
Processor, Memory, Calibration

### Cable Requirements

Power/Analog  
3-wire shielded cable  
Maximum distance is 13,300 feet with 14 AWG

### I/O Protection

Over-voltage, Miswiring, EMI/RFI Immunity



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# IECEx Certificate of Conformity

## INTERNATIONAL ELECTROTECHNICAL COMMISSION IEC Certification System for Explosive Atmospheres

for rules and details of the IECEx Scheme visit [www.iecex.com](http://www.iecex.com)

|                     |                                                                                                                        |             |                                      |
|---------------------|------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------|
| Certificate No.:    | <b>IECEx DEK 24.0039X</b>                                                                                              | Page 1 of 4 | <u>Certificate history:</u>          |
| Status:             | <b>Current</b>                                                                                                         | Issue No: 1 | <a href="#">Issue 0 (2025-01-29)</a> |
| Date of Issue:      | 2025-07-11                                                                                                             |             |                                      |
| Applicant:          | <b>Teledyne Gas &amp; Flame Detection</b><br>14880 Skinner Road<br>Cypress TX 77429<br><b>United States of America</b> |             |                                      |
| Equipment:          | <b>Toxic Gas or Oxygen Deficiency Sensor, VOC Gas Sensor</b>                                                           |             |                                      |
| Optional accessory: |                                                                                                                        |             |                                      |
| Type of Protection: | <b>Ex db ib</b>                                                                                                        |             |                                      |
| Marking:            | <b>Ex db ib IIC T4 Gb</b>                                                                                              |             |                                      |

Approved for issue on behalf of the IECEx  
Certification Body:

**R. Schuller**

Position:

**Certification Manager**

Signature:  
(for printed version)

Date:  
(for printed version)

2025-07-11

1. This certificate and schedule may only be reproduced in full.
2. This certificate is not transferable and remains the property of the issuing body.
3. The Status and authenticity of this certificate may be verified by visiting [www.iecex.com](http://www.iecex.com) or use of this QR Code.



Certificate issued by:

**DEKRA Certification B.V.**  
**Meander 1051**  
**6825 MJ Arnhem**  
**Netherlands**





# IECEx Certificate of Conformity

Certificate No.: **IECEx DEK 24.0039X**

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Date of issue: 2025-07-11

Issue No: 1

Manufacturer: **Teledyne Gas & Flame Detection**  
14880 Skinner Road  
Cypress TX 77429  
**United States of America**

Manufacturing  
locations: **Teledyne Detcon, Inc.**  
14880 Skinner Road  
Cypress, TX 77429  
**United States of America**

**Industrial Detection Solutions  
Company (IDS)**  
F11 & F12, Dammam 913 Street,  
Building No. 2857  
MODON-2  
Dammam Industrial City 2 34325  
**Saudi Arabia**

This certificate is issued as verification that a sample(s), representative of production, was assessed and tested and found to comply with the IEC Standard list below and that the manufacturer's quality system, relating to the Ex products covered by this certificate, was assessed and found to comply with the IECEx Quality system requirements. This certificate is granted subject to the conditions as set out in IECEx Scheme Rules, IECEx 02 and Operational Documents as amended

## STANDARDS :

The equipment and any acceptable variations to it specified in the schedule of this certificate and the identified documents, was found to comply with the following standards

**IEC 60079-0:2017** Explosive atmospheres - Part 0: Equipment - General requirements  
Edition:7.0

**IEC 60079-1:2014** Explosive atmospheres - Part 1: Equipment protection by flameproof enclosures "d"  
Edition:7.0

**IEC 60079-11:2011** Explosive atmospheres - Part 11: Equipment protection by intrinsic safety "i"  
Edition:6.0

This Certificate **does not** indicate compliance with safety and performance requirements other than those expressly included in the Standards listed above.

## TEST & ASSESSMENT REPORTS:

A sample(s) of the equipment listed has successfully met the examination and test requirements as recorded in:

Test Report:

**NL/DEK/ExTR24.0044/01**

Quality Assessment Reports:

**NL/DEK/QAR24.0012/00**

**US/ETL/QAR20.0010/05**



# IECEx Certificate of Conformity

Certificate No.: **IECEx DEK 24.0039X**

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Date of issue: 2025-07-11

Issue No: 1

## **EQUIPMENT:**

Equipment and systems covered by this Certificate are as follows:

DM-TOX Toxic Gas or Oxygen Deficiency Sensor, for fixed installation, designed to detect and monitor toxic gases or O<sub>2</sub> concentration in air.

Ambient temperature range -40 °C to +55 °C.

PI-VOC Gas Sensor, for fixed installation, designed to detect and monitor a wide range of VOC and toxic gases in air.

Ambient temperature range -20 °C to +55 °C.

## **SPECIFIC CONDITIONS OF USE: YES as shown below:**

1. The sensor shall be installed on a metallic grounded structure in a manner that the final assembly assures that the sensor enclosure is bonded to ground.
2. The sensor's common plane wire (black) shall be connected to ground in the final assembly.
3. Proper precautions shall be taken during installation, operation, maintenance and service of the product in order to avoid the build-up of static charges on the plastic splashguard.
4. The screws holding down the retaining plate are special fasteners of type: stainless steel, Philips Pan-head Machine screws, M3 x 0.5 6g/6H having yield strength greater than 40,000 PSI.
5. The flameproof joints are not intended to be repaired.



# IECEx Certificate of Conformity

Certificate No.: **IECEx DEK 24.0039X**

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Date of issue: 2025-07-11

Issue No: 1

**DETAILS OF CERTIFICATE CHANGES (for issues 1 and above)**

Update of the company name for the additional manufacturing location.